Stopping time and force

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain how a longer stop can reduce average force.

SUCCESS CRITERIA - I can...

1. Keep momentum change fixed in a fair comparison.
2. Compare force using impulse divided by time.
3. State why cushioning does not remove momentum change.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “impulse” and “interval” to explain your answer. Sentence starters are optional.

impulse: net force acting over time; change in momentum

interval: the time or range between two stated values

average: one value representing an interval or a set of values

01 Warm-up: bring back an earlier idea

A **3 kg** cart moving at **5 m/s** stops in **2 s**. **Find momentum-change magnitude** and average stopping-force magnitude.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

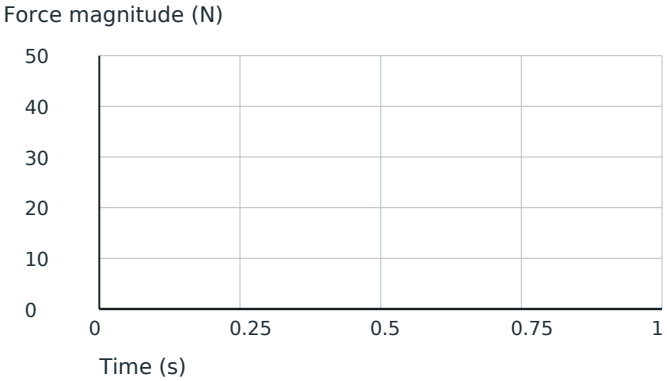
02 Worked example: explain the prediction

Reference: $|F_{avg}| = |\Delta p|/\Delta t$.
For the same momentum change, more stopping time lowers average force magnitude. Bars mean magnitude.

Worked reasoning: A stopping impulse has magnitude **8 N·s**. Over **0.2 s**, average force is **40 N**; over **0.8 s**, it is **10 N**. Longer stopping time lowers the average force for the same momentum change.

Your turn: Draw both force-time rectangles with equal areas.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .



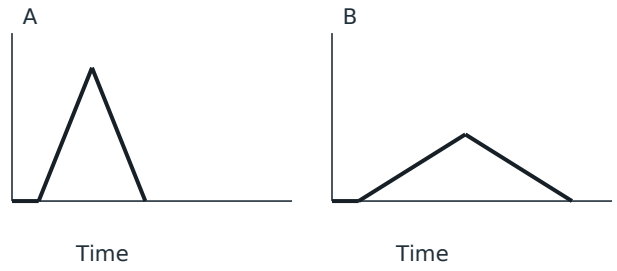
Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

The two ideal force pulses have equal area. **Name the longer-duration**, lower-average-force stop.

Word help: ideal = using stated simplifying conditions

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

A delivery crate loses **20 kg·m/s** of momentum as it stops. Packing A stops it in **1 s**; packing B stops it in **4 s**. **Complete the comparison table** and **find the average stopping-force magnitude** for each packing.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Stop	Time (s)	Impulse (N·s)	Average force (N)
A	____	____	____
B	____	____	____

Marzano check - circle: ① ② ③

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Continue down each column.

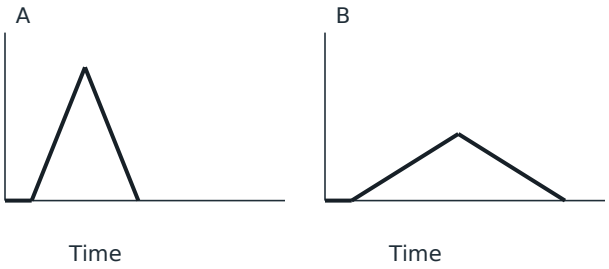
Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a cushion makes momentum disappear. **Explain where the momentum change** comes from and what cushioning changes.

Sentence starter: I think … . I can point to …, which shows …, so … .

Force (same scales)



Same scales: compare impulse and peak force.

Marzano check - circle: ① ② ③

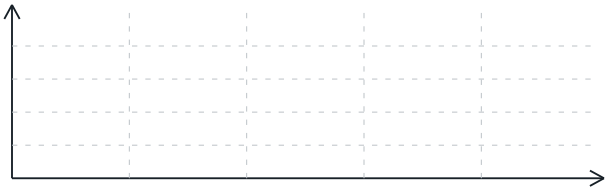
06 Review: connect an earlier idea

Downward velocity changes from **4** to **10 m/s** in **0.6 s**. **Find acceleration** and **name one possible measurement uncertainty**.
Sketch the motion and **label a positive direction**.

Word help: acceleration = rate of change of velocity

Sentence starter: I found … [unit]. This value means … in this situation.

Velocity (m/s)



Marzano check - circle: ① ② ③

07 Exit check

The same object stops from the same speed in twice as much time. **Compare average force sizes**.

Sentence starter: Both … . They differ in … because … .

Trial	Duration	Average force
Original	Δt	F
New	____	____

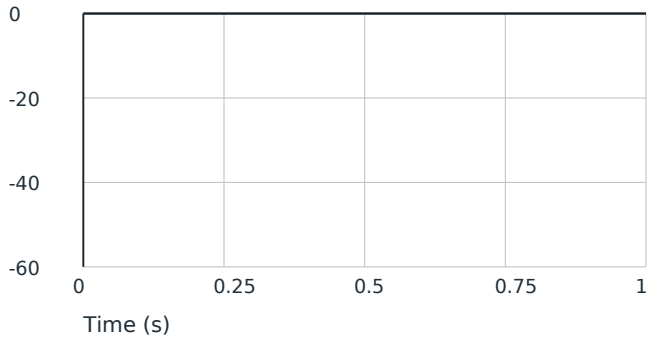
Marzano check - circle: ① ② ③

08 STRETCH: test the prediction

A cushion doubles stopping time without changing incoming or final velocity. **Compare impulse** and average force; **state what cannot be concluded** about the exact peak force.

Sentence starter: Both … . They differ in … because … .

Stopping force (N)



Marzano check - circle: ① ② ③